

Noise-Aware Transformer-Based Low-Light Image Enhancement with Downstream Object Detection Evaluation

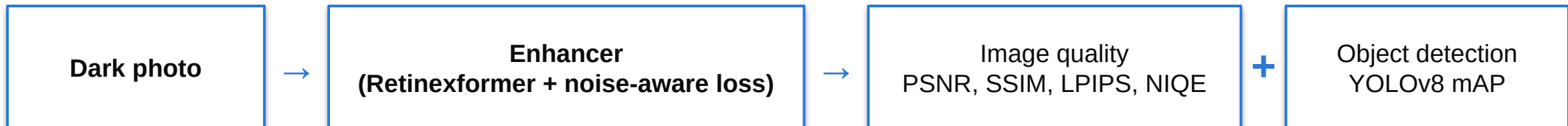
Mid-semester B.Tech Project evaluation

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Project at a glance: what the title means

Goal: make dark photos bright and clean - without amplifying noise - and check whether this helps a machine see.

Transformer-based enhancement	Noise-aware	Downstream detection evaluation
Retinexformer (ICCV 2023) as the main model Reproduce it and compare with 8 baselines under one protocol (quality, no-reference, speed) Reproduced: 25.15 dB vs paper 25.16 → pipeline verified	SNR-weighted training loss SNR map finds dark, noisy regions; the training loss gives them more weight; compared with a control run Pilot tested; next: noise awareness inside the network	Does enhancement help a detector? YOLOv8m on ExDark (1,200 dark images, 12 classes): raw images vs every enhanced version Result: raw images best (mAP50 0.662); no enhancer helps



Why low-light enhancement?

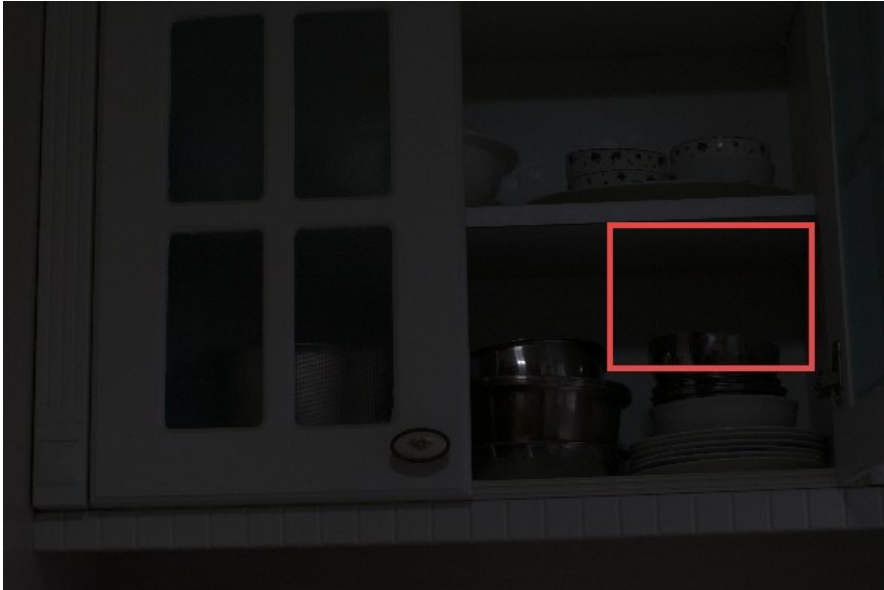


Our run: LOL-v1 test image 79 - dark input (left), Retinexformer output (right)

Where dark images occur:

- Phone night photography
- Surveillance cameras
- Night driving, robots, drones
- Detectors miss objects in the dark

The problem: darkness + noise + colour shift



Our LOL-v1 test image 79; red box = dark region zoomed on the right

Input $\times 6$ (just brighter)



Zero-DCE



Ground truth



- Dark: little light reaches the sensor, details are hidden
- Noisy: few photons $\rightarrow$ low SNR (signal-to-noise ratio) $\rightarrow$ speckles
- Colour shift: weak light distorts colours (grey-green instead of white)

Brightening multiplies noise as much as signal $\rightarrow$ enhancement must brighten AND denoise

How the field evolved



- Hand-made rules → learned from data → no pairs needed → long-range attention → generative
- Survey (Li et al., TPAMI 2022): no method wins everywhere; noise in dark regions remains open
- This project: Retinexformer (ICCV 2023) as the main model

Key methods compared in this project

Method	Year / venue	Key idea	Limitation
HE / CLAHE / Gamma	classical	Re-map brightness	No noise model, colour shifts
Zero-DCE	CVPR 2020	Learned brightness curves, no pairs	Brightens noise
SCI	CVPR 2022	Self-calibrated illumination, ~300 params	Brightens noise
SNR-Aware	CVPR 2022	SNR map decides local vs. long-range	Heavier; no Retinex model
LLFormer	AAAI 2023	Axis-based attention for 4K/8K images	24.5 M params, slower
Retinexformer	ICCV 2023	Illumination-guided channel attention	Illumination-aware, not noise-aware
GSAD	NeurIPS 2023	Diffusion with structure regularization	Slow; GT-brightness test trick

Research gap and objectives of this midsem

Gaps

- Noise amplification in dark regions is unsolved
- Retinexformer: illumination-aware, not noise-aware
- Methods judged by PSNR/SSIM, rarely by a machine task
- Diffusion: quality at a large speed cost

Objectives (first half)

- Reproduce Retinexformer + 8 baselines
- One protocol: PSNR, SSIM, LPIPS, NIQE, dark regions, speed
- Detection study: does enhancement help YOLOv8?
- Noise-aware SNR-weighted loss, tested against a control
- Plan the second half from the evidence

Background: Retinex theory and attention cost

Retinex model

$$I = R \odot L$$

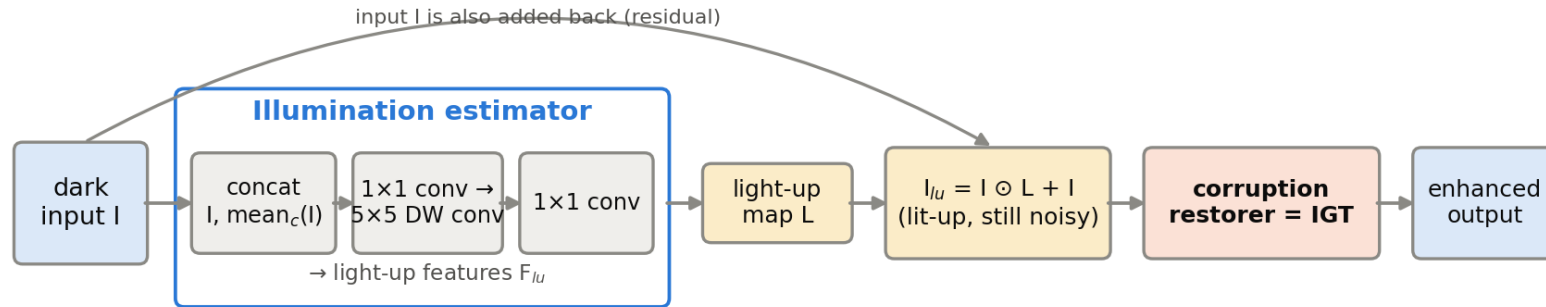
- I : observed image
- R : reflectance = object colours
- L : illumination = light on the scene
- Enhance: estimate L , replace by brighter light
- Real images: noise in R and L , amplified when L is small

Why attention is costly

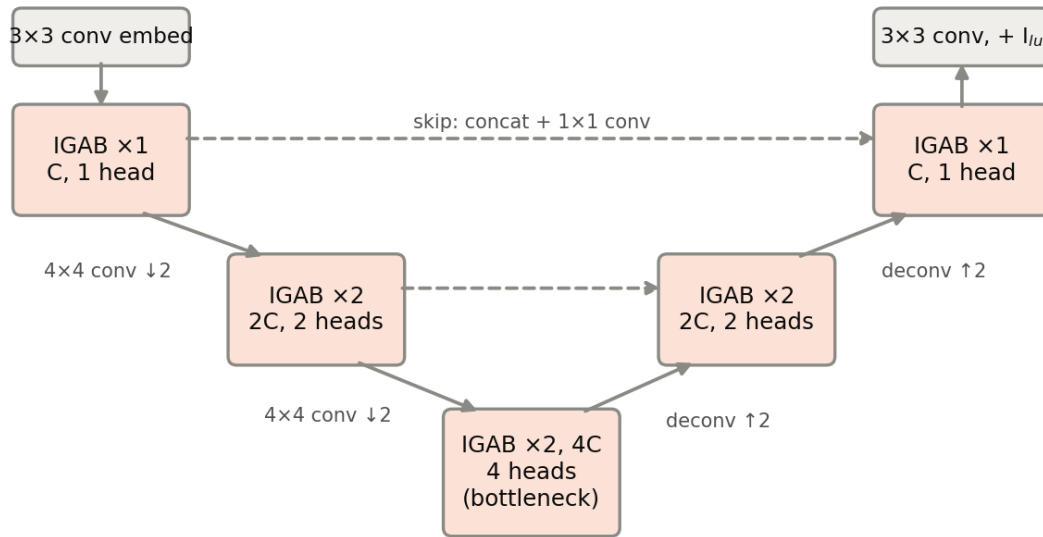
$$\text{softmax}(QK^T/\sqrt{d}) \cdot V$$

- QK^T is $N \times N$ (N = number of pixels)
- 600×400 image: $N = 240,000 \rightarrow 58$ billion entries
- Channel attention: $\text{softmax}(K^TQ)$ is $C \times C$
- $\rightarrow$ cost grows linearly with image size

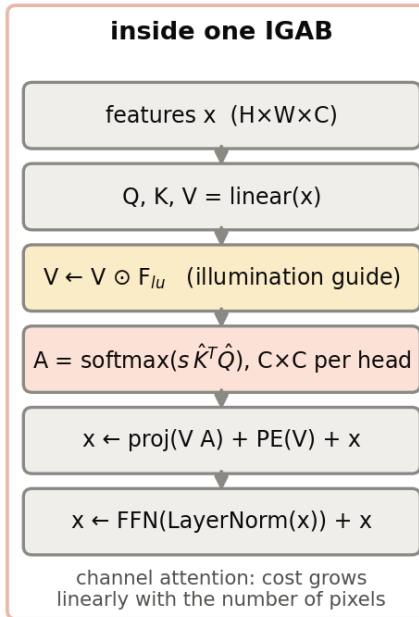
Main model: Retinexformer (ICCV 2023)



Illumination-Guided Transformer (IGT), C = 40 channels

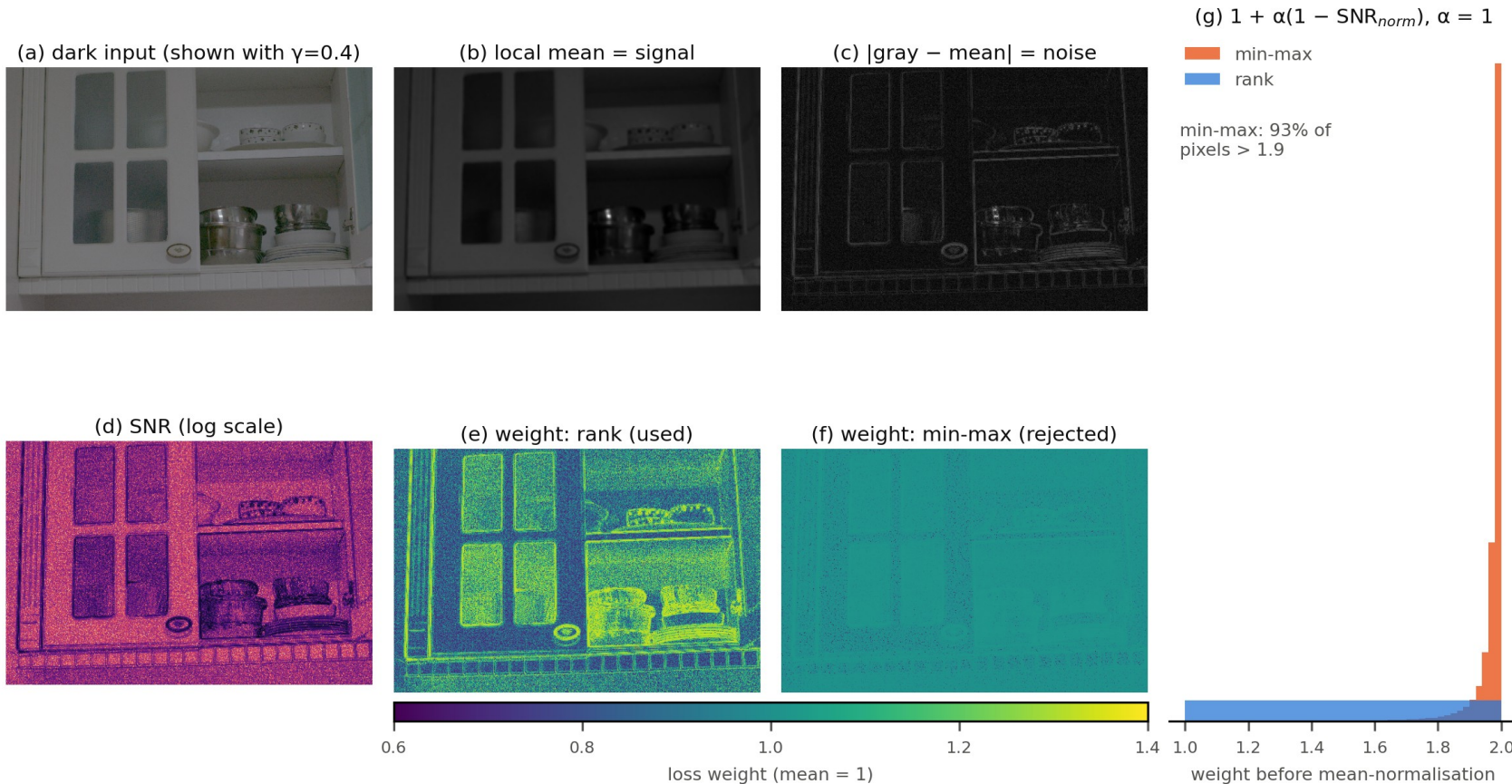


F_{Iu} is down-sampled together with the features and fed to every IGAB.



- Estimator → light-up map L
- Lit-up image: $I \odot L + I$ (still noisy)
- Restorer: U-shaped transformer (IGT)
- IG-MSA: $V \times$ light-up features, attention across channels
- 1.61 M params, ℓ_1 loss

My idea: noise-aware (SNR-weighted) loss



Steps

- blur $b = 5 \times 5$ mean of gray
- noise $n = |\text{gray} - b|$
- $\text{SNR} = b / (n + \epsilon)$
- rank $r \in [0, 1]$ per image
- $w = 1 + \alpha(1 - r)$, $\alpha = 1$
- $w \leftarrow w / \text{mean}(w)$
- $L = \text{mean}(w \cdot |\hat{Y} - Y|)$

Rank normalisation (e) spreads the weights; min-max (f) gives almost every pixel the same weight (93% > 1.9).

Experimental setup

Data	Use	Test images
LOL-v1	paired, main	15
LOL-v2 real / syn	paired	100 / 100
LIME / DICM / MEF	unpaired, NIQE	10 / 69 / 17
ExDark (12 classes)	detection	1,200 (200 for GSAD)

Metric	Measures	Better
PSNR	pixel fidelity	↑
SSIM	structure	↑
LPIPS	perceptual distance	↓
NIQE	naturalness (no GT)	↓
Dark-30% PSNR	darkest pixels only	↑
mAP50	detection accuracy	↑

- Hardware: Kaggle Tesla T4 (training, timing); laptop for evaluation
- Rules: no GT at test time · no test-chosen checkpoint · no train/test overlap

Quantitative results (paired test sets)

Method	LOL-v1 PSNR↑	SSIM↑	LPIPS↓	LOL-v2-real PSNR↑	SSIM↑	LPIPS↓
Input	7.77	0.191	0.560	9.72	0.196	0.519
Gamma	14.53	0.620	0.351	18.70	0.659	0.318
Zero-DCE	14.86	0.562	0.335	18.06	0.580	0.313
SCI	14.85	0.527	0.339	17.36	0.542	0.307
SNR-Aware*	24.61	0.840	0.151	21.48	0.848	0.157
LLFormer	23.65	0.816	0.169	—	—	—
GSAD	22.73	0.850	0.103	20.14	0.845	0.114
Retinexformer	25.15	0.843	0.131	22.79	0.839	0.171
Retinexformer (mine)	23.10	0.828	0.143	—	—	—

* images released by the SNR-Aware authors. Best value in bold. —: train/test overlap (see setup).

Reproduction and protocol findings

Method / set	Paper	Ours
Retinexformer, LOL-v1	25.16	25.15
SNR-Aware, LOL-v1	24.61	24.61
LLFormer, LOL-v1	23.65	23.65
GSAD, LOL-v1 (GT trick)	27.84	27.57
GSAD, LOL-v1 (real output)	—	22.73

PSNR in dB on LOL-v1

- GSAD's GT-brightness trick: +4.85 dB on LOL-v1
- LOL-v2-real test: 91 / 100 images are LOL-v1 training images
- My re-training: 23.10 dB vs released 25.15 (others: 23.45)

Qualitative results



Does enhancement help a detector? (ExDark, YOLOv8m)

Input to YOLOv8m	mAP50 (1,200)	Recall	mAP50 (200)
Raw (no enhancement)	0.662	0.611	0.707
Retinexformer	0.603	0.540	0.628
SCI	0.595	0.529	0.625
Zero-DCE	0.572	0.502	0.603
GSAD	—	—	0.633

- Raw images give the best mAP
- Enhancement lowers recall, not precision
- Better restoration hurts less, none helps



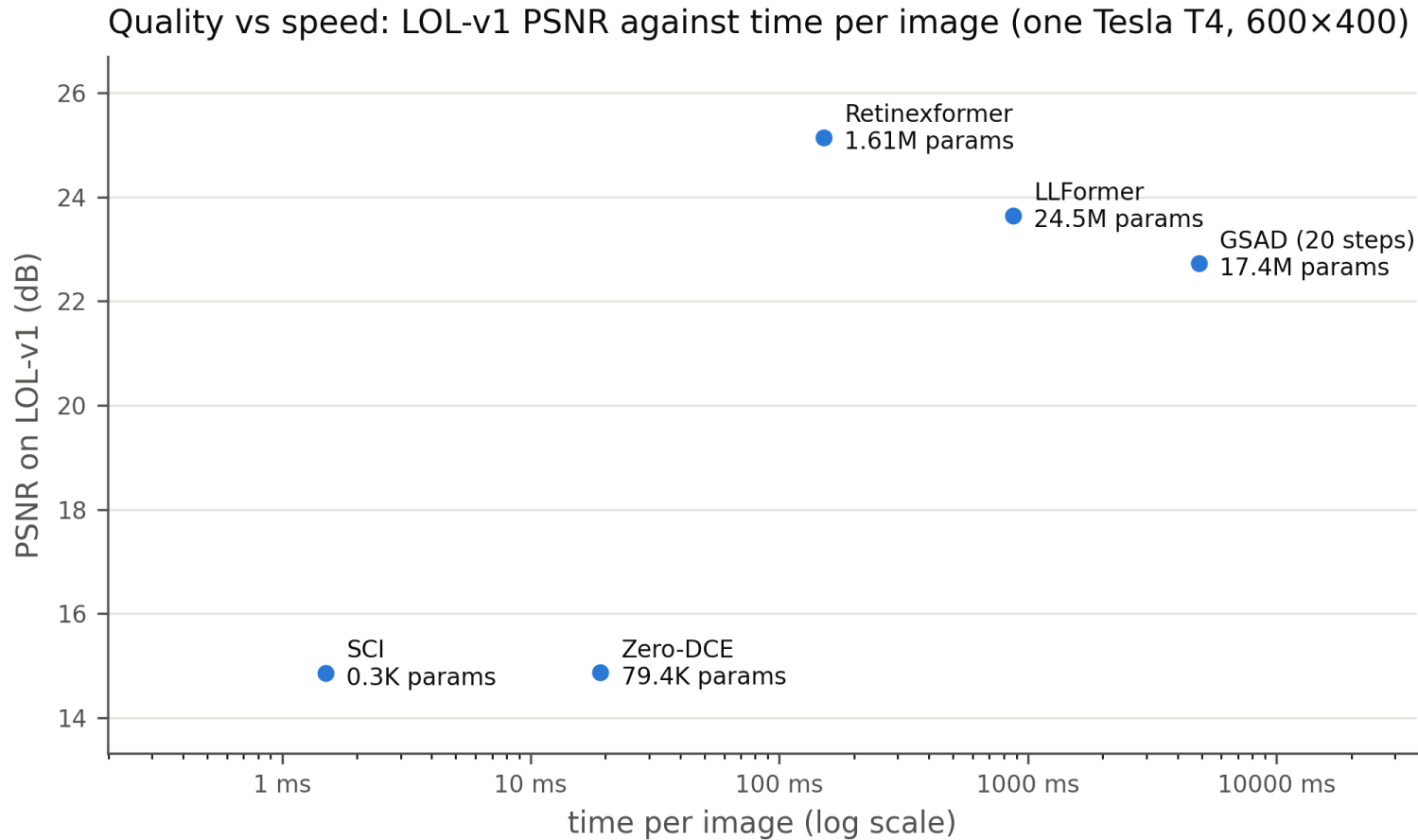
Ablation: does the noise-aware loss help?

Run	LOL-v1 PSNR	Dark-30% PSNR	Dark wins vs A	ExDark mAP50
Released (start)	25.15	25.31	–	0.627
A: L1 (control)	24.31	24.89	–	0.627
B: SNR-L1 (mine)	24.32	24.91	13/15	0.626
C: L1 + FFT	24.31	24.89	3/15	0.626
D: SNR-L1 + FFT	24.32	24.91	12/15	0.626

- B vs A: dark regions better on 13/15 images ($p = 0.004$)
- ...but only +0.01 to +0.05 dB: negligible
- FFT loss: no effect
- Fine-tuning lowers LOL-v1 PSNR for every run

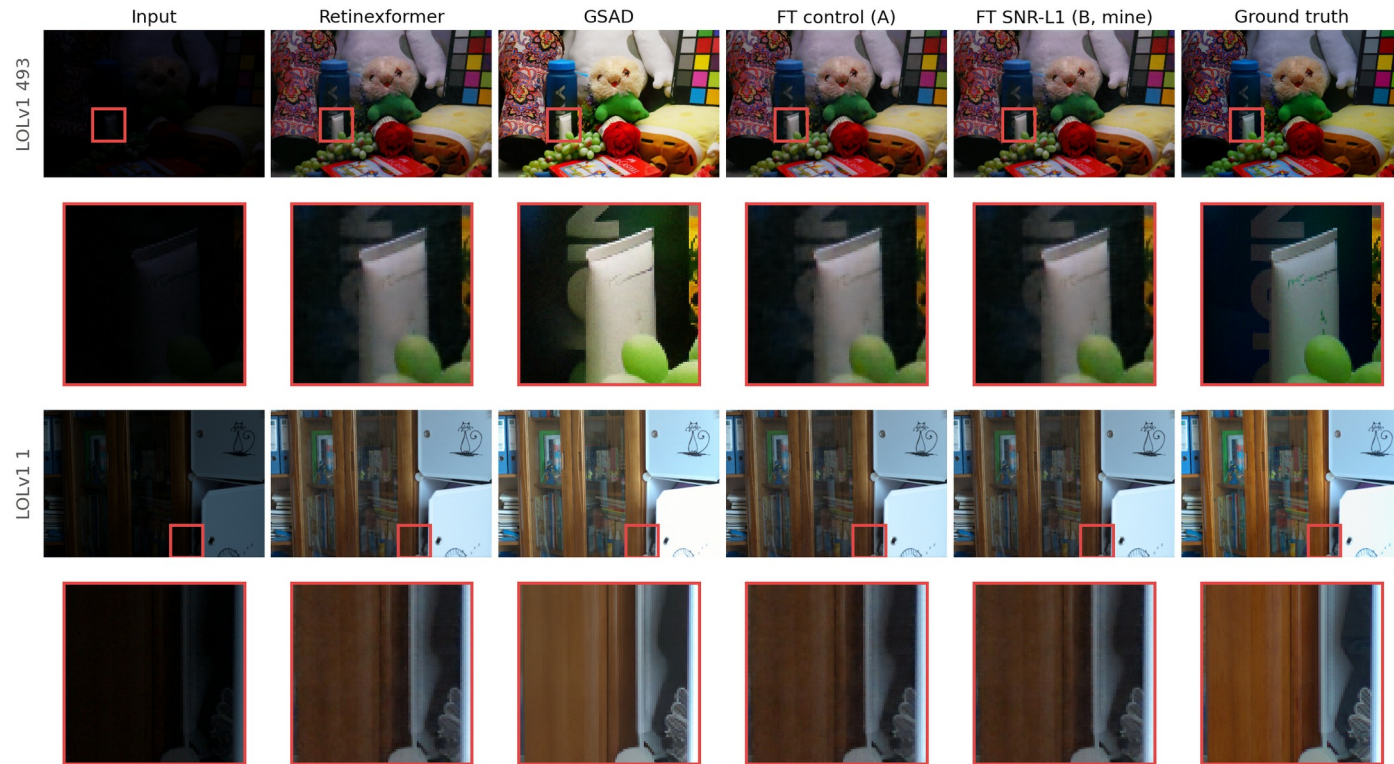
Takeaway: a consistent effect in the intended place (dark regions), but far too small to matter. The idea needs to act inside the network, not only in the loss.

Efficiency: quality vs. speed (one Tesla T4)



- SCI: 1.5 ms, low quality
- Retinexformer: 151 ms, best PSNR
- LLFormer: 874 ms, 24.5 M params
- GSAD: 2.4–4.9 s per image

Limitations and failure cases



- Only 15 LOL-v1 test images
- One seed per run; limited free GPU time
- SNR estimate also marks edges as noise
- Detector used as-is (not fine-tuned)
- SNR-Aware: released images only; GSAD on 200 ExDark images

Plan for the second half

Weeks 1–2

SNR guidance inside IG-MSA; noise estimate that ignores edges

Weeks 3–4

Sweep α and weighting; 3 seeds per setting

Weeks 5–6

Train best variant from scratch; larger test sets

Week 7

Fine-tune YOLOv8 on enhanced vs raw ExDark

Week 8

Final report and code release

Conclusions

- Reproduced Retinexformer (25.15 dB) + 8 baselines under one protocol
- Found: GT-brightness trick (GSAD), LOL-v2-real / LOL-v1 overlap
- Raw images beat every enhancer for YOLOv8m (mAP50 0.662)
- SNR-weighted loss: consistent but negligible gain in dark regions
- Next: noise awareness inside attention, larger tests

Key references: Cai et al., Retinexformer, ICCV 2023 · Xu et al., SNR-Aware, CVPR 2022 · Hou et al., GSAD, NeurIPS 2023 · Wang et al., LLFormer, AAAI 2023 · Guo et al., Zero-DCE, CVPR 2020 · Ma et al., SCI, CVPR 2022 · Li et al., LLIE survey, TPAMI 2022 · Loh & Chan, ExDark, CVIU 2019 (full list in the report)